Tautification and the shifted nilcharacter construction

Project proof draft: /root/astra_v14_baseline_theta

7 November 2026

Status. This is an author proof proposal. Its internal geometric input is the rational-subgroup and quotient argument in the separately frozen local Lemma 13.2 proof [2], which is also awaiting independent acceptance. The present unit does not assert that the complete inverse theorem has been repaired.

1 The exact change of filtration

All Lie groups and rationality conventions are those of [1, Sections 3–6]. A rational subgroup is connected, and its intersection with the ambient lattice is cocompact. The three domain filtrations used here are the degree filtration on $\mathbb{Z}^a$, the multidegree filtration on $\mathbb{Z}^a$, and the degree-rank filtration on $\mathbb{Z}$ from [1, Definition 6.17]. For limit objects, defining relations are interpreted by their natural ultrapower extension as in [1, Appendix A.2–A.3, pp. 1324–1325]. Thus vertical equivariance quantifies over every limit point of the nilmanifold and every limit element of the top group, using the extension of the standard vertical character. We do not infer this exact internal identity from an identity asserted only at standard points. For these three cases respectively put

$$H = G_1, \qquad H = \langle G_{e_1}, \dots, G_{e_a} \rangle_{\mathbb{R}}, \qquad H = G_{(1,0)}.$$

In the middle case the connected real generated subgroup equals the group generated by the connected subgroups: the latter is the connected analytic subgroup with the generated Lie algebra, and such subgroups in a simply connected nilpotent group are closed. The rational-subgroup lemma in [2, Section 2] shows that H is rational.

Theorem 1. *Suppose G/Γ is a rational filtered nilmanifold with finite downset of degrees J for one of these three domain filtrations. Every polynomial nilsequence $F(g(n)\Gamma)$ admits an exactly equal representation on $H/(H \cap \Gamma)$ with filtration*

$$G'_0 = H, \qquad G'_i = G_i \quad (i > 0).$$

For degree-rank, 0 denotes $(0,0)$. This filtration is taut and has the same positive-degree groups and no larger degree support, hence the same allowed degree bound. Its test-function Lipschitz bound depends only on the original fixed filtered nilmanifold and original Lipschitz bound, not on the polynomial coefficients. For a nilcharacter of positive degree d, the same standard vertical character on G_d is preserved. These assertions hold uniformly for limit maps and families of polynomial coefficients with fixed geometric data.

Proof. As in [2, Section 3], first restrict to G_0 and its lattice if necessary. Choose $g(0) = q\gamma$ with q in a fixed compact representative set, and set

$$h(n) = q^{-1} g(n) \gamma^{-1}, \qquad F_q(x\Gamma) = F(qx\Gamma).$$

Then $h(0) = 1$, $F_q(h(n)\Gamma) = F(g(n)\Gamma)$, and h is polynomial for the old filtration. Indeed every G_i is normal in G_0; use the constant map $g(0)^{-1}$, product closure in [1, Corollary B.4], and filtered conjugation by q. The Lipschitz bound is uniform because q lies in a fixed compact set.

We show that $h(n)$ belongs to H. In the degree case the difference $h(n)h(0)^{-1}$ is in G_1 for every $n \in \mathbb{Z}^a$, by the derivative condition. In the degree-rank case the same statement holds with $G_{(1,0)}$. In the multidegree case connect 0 to n by a finite path of positive or negative coordinate steps. At a step in the jth coordinate, $h(v \pm e_j)h(v)^{-1} \in G_{e_j}$. Multiplying these increments along the path and using $h(0) = 1$ proves the assertion. This argument applies to every integer input and puts no bound on its size.

Every original positive group is contained in H. For degree and degree-rank this follows from nesting and the least positive index. For multidegree, any positive index dominates some e_j. Consequently the displayed new groups are well-defined subgroups of H, are descending, and satisfy the commutator axioms inherited from G, including those with index zero because $H \subseteq G_0$. Their intersections with $H \cap \Gamma$ equal their intersections with Γ and are lattices. In degree-rank all the required equalities $G_{(d,0)} = G_{(d,1)}$ for $d > 0$ remain unchanged. The new group at zero is exactly the required positive group or group generated by the minimal positive groups, so the new filtration is taut.

All positive iterated derivatives of h have the same required group membership as before, and the zeroth-order membership was just checked. Thus h is polynomial for the new filtration. Restrict F_q through the fixed smooth inclusion $H/(H \cap \Gamma) \longrightarrow G/\Gamma$. The inclusion is an embedding of compact manifolds, so this costs only a fixed Lipschitz comparison factor and gives the asserted exact sequence.

For a nilcharacter let $\eta : G_d \longrightarrow \mathbb{R}$ be its standard vertical character. The proof in [2, Section 3] applies verbatim with G_d in place of its top cube group: for each $t \in G_d \cap \Gamma$, the continuous map $q \mapsto \eta(qtq^{-1})$ on connected G_0 is integer-valued by the unit-norm vertical identity at $q\Gamma$. It is therefore constant. The lattice spans G_d as a connected real group, so $\eta(qtq^{-1}) = \eta(t)$ for all $t \in G_d$ and $q \in G_0$. Hence F_q and its restriction have the same standard vertical character.

For families, choose one bounded standard representative section of the fixed compact quotient. Such a section can be formed from finitely many local smooth sections, taking the first available chart. Apply this one section to each constant coefficient. The resulting choices are limit maps when the coefficients are limit maps. Discontinuities in the parameter do not enter a Lipschitz bound in the orbit variable. All metric constants and groups remain fixed. The exact identities can be imposed on representatives on an ultrafilter-large set by the witness argument of [2, Section 7]; the ordinary proof then passes to the limit. □

This also proves equality of the classes of nilsequences under the old and taut conventions, at each finite degree support: one inclusion is the theorem and the reverse inclusion is immediate. It therefore preserves the equivalence relation defined by a lower-degree nilsequence. It is not merely an assertion that the top-degree function has a similar replacement.

Corollary 2. *In a taut degree or degree-rank filtration of degree at most (d, r) or d, the ambient group has nilpotency class at most d. In a taut multidegree filtration supported on a finite downset J, its class is at most $D = \max_{a \in J} |a|$. If the support is contained in the downset below one positive index d, the group G_d is central.*

Proof. For degree and degree-rank, brackets of j elements of the ambient first positive group belong respectively to G_j or $G_{(j,0)}$, which vanish when $j > d$. For multidegree, expand an iterated Lie bracket using the generating Lie algebras $\mathfrak{g}_{e_i}$. A bracket of length j lies in some $\mathfrak{g}_a$ with $|a| = j$, hence vanishes for $j > D$. The generated Lie algebra has the asserted nilpotency class, and so

does its simply connected group. Finally $[G_d, G_{e_i}] \subseteq G_{d+e_i} = 1$ in the multidegree case. The same argument uses the first positive group in the other two cases. Since these groups generate the ambient group, G_d is central. □

2 A complete shifted nilcharacter construction

Theorem 3. *For any of the three domain filtrations above, let χ be a nilcharacter of positive degree d. Let $i > 0$ and let u belong to the domain group at index i. Then*

$$n \longmapsto \chi(n+u) \otimes \overline{\chi(n)}$$

is a nilsequence with support strictly below d. The representing geometric data and Lipschitz bound can be fixed independently of u within this domain group. For limit u the representation and test function can be chosen in a limit fashion. This supplies the shift assertion required in [1, Lemma E.8(iv)] for the three domains used in the proof.

Proof. By Theorem 1 represent the same χ on a taut nilmanifold G/Γ, with the same degree and vertical character. Write its polynomial map as g. In the degree-rank case use $i = (1, 0)$: any nonzero element of a positive domain group requires this index, and if the given domain group is trivial then $u = 0$ also belongs to this first group. Put $U(v) = (v, 1)$ and $D(v) = (v, v)$ in G^2. For each index j let L_j be the connected subgroup generated by

$$U(G_{j+i}) \quad \text{and} \quad D(G_j).$$

These groups are rational by [2, Lemma 2], and are descending. The generator commutators have the following memberships:

$$\begin{aligned} [D(G_j), D(G_k)] &\subseteq D(G_{j+k}), \\ [D(G_j), U(G_{k+i})] &\subseteq U(G_{j+k+i}), \\ [U(G_{j+i}), U(G_{k+i})] &\subseteq U(G_{j+k+2i}) \subseteq U(G_{j+k+i}). \end{aligned}$$

Passing to generated Lie algebras and using BCH, as in [2, Section 4], proves $[L_j, L_k] \subseteq L_{j+k}$, including index zero. Thus L_0/Δ, with $\Delta = L_0 \cap \Gamma^2$, is a rational filtered nilmanifold of degree at most d. In the degree-rank case its additional equalities hold: $L_{(a,0)} = L_{(a,1)}$ follows from $G_{(a,0)} = G_{(a,1)}$ and $G_{(a+1,0)} = G_{(a+1,1)}$ for $a > 0$.

The difference $a_u(n) = g(n+u)g(n)^{-1}$ is polynomial for the shifted filtration $(G_{j+i})_j$. Its zeroth-order value belongs to G_i. Every further iterated derivative is an iterated derivative of g with the derivative in direction u innermost, so the defining index sum has the additional term i. Therefore $U(a_u(n))$ and $D(g(n))$ are polynomial for (L_j). By [1, Corollary B.4], so is their product

$$P_u(n) = (g(n+u), g(n)).$$

This verifies the actual orbit, not only its coordinate projections.

As $i > 0$, $G_{d+i} = 1$ and $L_d = D(G_d)$. Corollary 2 shows that G_d is central in G_0, hence L_d is central in L_0. The function

$$T(x\Gamma, y\Gamma) = F(x\Gamma) \otimes \overline{F(y\Gamma)}$$

restricted to L_0/Δ is invariant under $D(G_d)$, since its two vertical characters cancel. The rational central-quotient argument in [2, Sections 2 and 6] therefore gives a nilsequence on L_0/L_d with the exact value $T(P_u(n)\Delta)$. Its filtration group at d is trivial and all groups outside the downset below

d were already trivial. Its support is consequently strictly below d. The same argument gives a fixed Lipschitz bound on the quotient.

The quotient may not yet be taut. Apply Theorem 1 to this lower-degree nilsequence, preserving its exact values without enlarging its degree support. The groups L_j and their quotient depend on the index i but not on the element u. The map $u \mapsto P_u(0)$ is a limit map when g and u are limit data. The one fixed bounded section used in Theorem 1 then gives a limit choice of the final normalized representation, with uniform Lipschitz bounds independent of u. This proves all the assertions. □

For the erratum's needed case the domain is $\mathbb{Z}^2$ with multidegree, $d = (1, s-1)$, $i = (1, 0)$ and $u = (k, 0)$. The resulting support is contained in

$$\{a : a \leq (0, s-1)\} \ \cup \ \{a : a \leq (1, s-2)\},$$

uniformly in k. This is exactly the support needed before applying the published splitting and vertical Fourier lemmas in the proof of the new difference decomposition. Those applications, and the subsequent two Cauchy–Schwarz steps, are separate proof obligations.

3 Scope of the correction

The tautification theorem supplies the general representation repair suggested, but not fully proved, in Section 2 of the author erratum. It also proves the top centrality and nilpotency-class assertions for the taut conventions. The shifted construction above supplies the needed instance of E.8(iv) without leaving its auxiliary rationality or central quotient implicit. This unit does not claim to have audited every filtration introduced in the published proof. In particular, E.8(v), E.9, Lemma 12.1, the Section 12 quotient, and the final modified Proposition 7.3 argument still need their own exact applications or replacements.

References

[1] B. Green, T. Tao and T. Ziegler, *An inverse theorem for the Gowers $U^{s+1}[N]$-norm*, Annals of Mathematics **176** (2012), 1231–1372.

[2] Project proof, *A local replacement for the filtration argument in GTZ Lemma 13.2*, draft 002, 7 November 2026. Manifest SHA-256:

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This is an explicitly linked author proof awaiting independent acceptance.